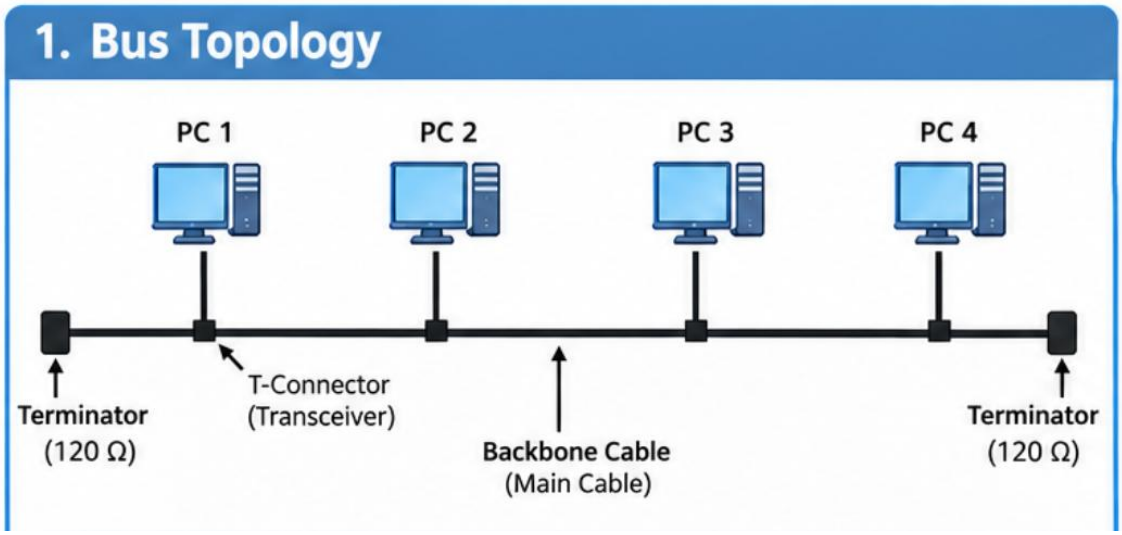
# Assignment – 1.18

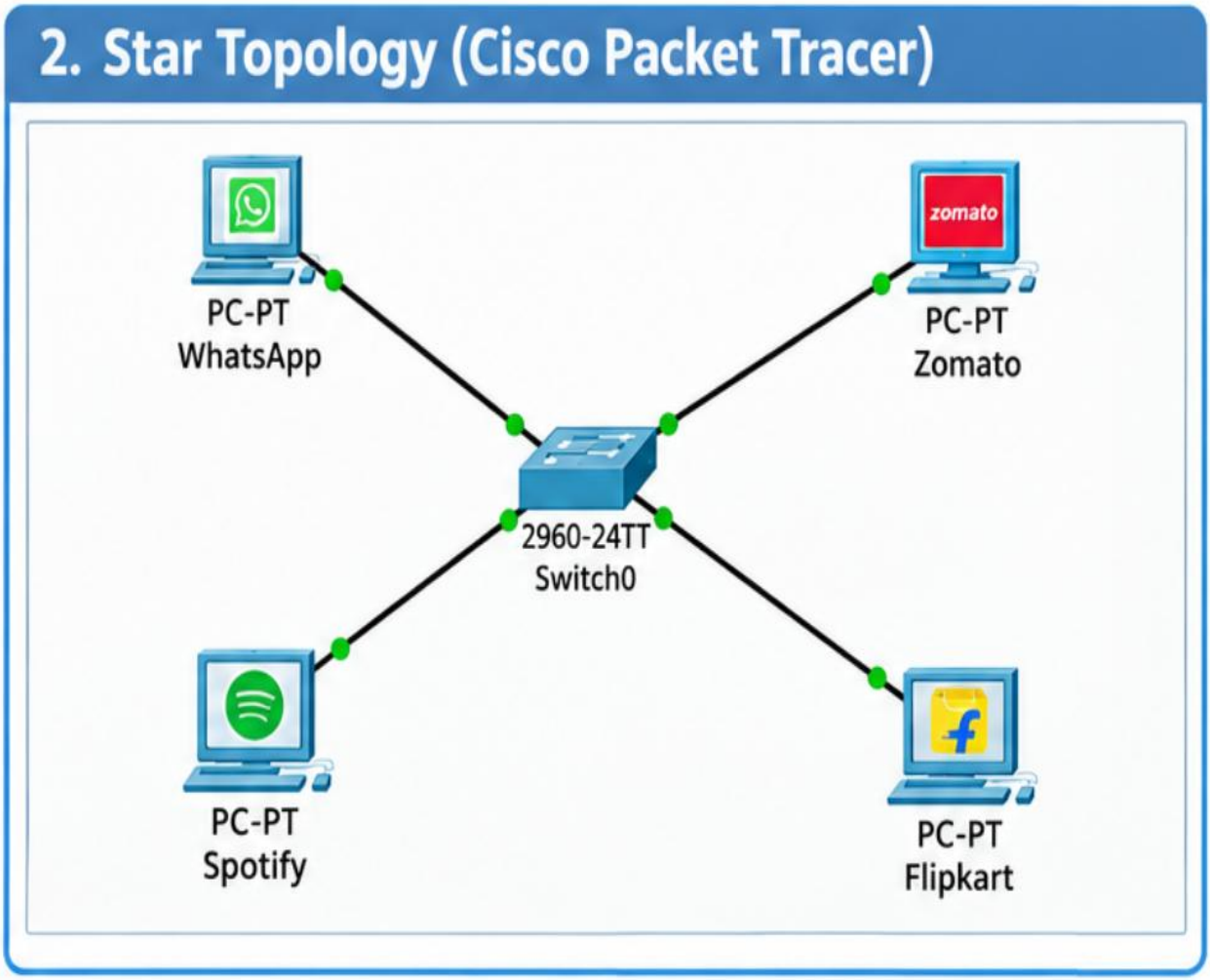
# Network Topologies Theory

**Q - 1. Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.**

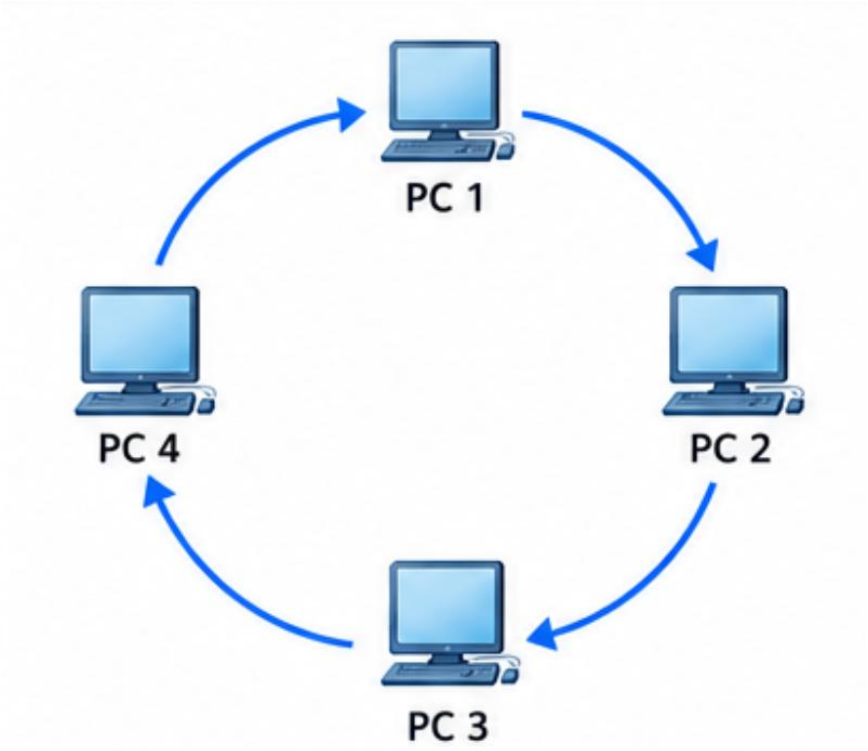



- **Real-world example:**
  Older Ethernet networks, such as legacy networks in factories and older office buildings, sometimes used bus topology.

**<u>Q - 2.</u> In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.**



**<u>Q - 3.</u> Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup. Hint: Use arrows to show the direction of data flow.**




- **<u>How data travels:</u>**

  In a ring topology, data travels in one direction from device to device, passing through each node in the ring until it reaches the destination. Each device forwards the data to the next device in the circle.

## Q - 4. Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

- **Mesh vs. Star Topology:**

| Topology | Advantages | Disadvantages |
|---|---|---|
| **Mesh** | 1. Very reliable.<br>2. Multiple paths for data. | 1. Expensive.<br>2. Difficult to set up. |
| **Star** | 1. Easy to install.<br>2. Easy to manage. | 1. Switch failure affects all PCs.<br>2. Needs more cables than bus topology. |